

AMENDMENT NO. 2 AUGUST 1990
TO
IS 9974 (Part 2) : 1981 SPECIFICATION FOR HIGH
PRESSURE SODIUM VAPOUR LAMPS
PART 2 STANDARD LAMP DATA SHEETS

(Page 4, clause 3.1) — Insert the following at the end:

- 'g) 70 W Tubular clear
- h) 70 W Elliptical bulb — diffuse coating clear'

(Page 22, Technical Data Sheet No. 6) — Insert the following data sheets No. 7 and 8 at the end.

TECHNICAL DATA SHEET NO. 7
HIGH PRESSURE SODIUM VAPOUR LAMPS

RATED WATTAGE : 70 W FOR USE WITH EXTERNAL IGNITOR TUBULAR BULB — CLEAR

LAMP STARTING TEST

Test voltage (V)	198
Maximum starting time (s)	10

PULSE CHARACTERISTICS [see FIG. 1 OF IS : 9974 (Part 1) - 1981]

Height (V)	$1\,775 \pm 25^*$
Waveshape	Sinusoidal*
Direction	A positive pulse during the positive half cycle and a negative pulse during the negative half cycle of the rms voltage wave†
Position	90 and 270 electrical degrees of open circuit voltage
Rise time — T_1 Max	$1.00\ \mu s^*$
Duration time — T_1	$1.95 \pm 0.05\ \mu s^*$
Repetition rate	One per each half cycle

*These values are based on the assumption that this sinusoidal pulse will give starting results equivalent to a rectangular pulse of $1\,775 \pm 25$ V, very short rise time and a duration time of $1.95 \pm 0.05\ \mu s$. This matter is under investigation together with further details concerning the measuring circuit.

†The starting device used for the specified lamp starting test shall supply a pulse current of 1.8 ± 0.2 A and shall not allow pulse current reversal.

LAMP WARM-UP TEST

Test voltage (V)	198
Maximum time required to reach 50 V minimum at lamp terminals (Min)	7

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LAMP ELECTRICAL CHARACTERISTICS (AT RATED VOLTAGE OF REFERENCE BALLAST)			
	<i>Objective</i>	<i>Maximum</i>	<i>Minimum</i>
Voltage at lamp terminals (V) (rms)	90	105	75
Lamp current (A) (rms)	0.98	—	—
Lamp wattage (W)	70	—	—
Extinguishing voltage at lamp terminals (V) (rms)	105	—	—
Rated luminous flux (lumens)	5 200	—	—

REFERENCE BALLAST CHARACTERISTICS

Rated frequency (Hz)	50
Rated voltage (V)	220
Calibration current (A)	0.98
Voltage/current ratio	188
Power factor	0.075 ± 0.005

LAMP DIMENSIONS (see FIG. 1)

Cap	Bulb Diameter (Max) <i>D</i>	Overall Length (Max) <i>L</i>	Light Centre Length <i>C</i>	Arc Length <i>A</i>	Deviation of Any Point Along Centre Line of Arc Tube from Axis of Cap (Apex of Cap Eyelet Used as the Point of Reference)	Operating Position Limita- tion
(1)	(2) mm	(3) mm	(4) mm	(5) mm	(6)	(7)
E27	38.5	156	105 ± 10	28.45	—	As indicated by lamp manufac- turer

BALLAST DESIGN INFORMATION [see B OF IS : 9974 (Part 1) - 1981]

	<i>Maximum</i>	<i>Minimum</i>
Lamp warm-up current for ballast design (A) (rms)	1.96	0.98
Pulse height for ballast design (V)	2 300	see Note

Note — There are two designs of lamp presently in use which are compatible in operation, but which require different starting conditions.

Some lamp designs require a minimum pulse height of 1 600 V whereas other designs require a minimum pulse height of 1 800 V. Information is to be provided by the lamp manufacturer as to which ignitor pulse height and width are appropriate.

For future starting compatibility of both lamp types, it is recommended that ignitors are designed with a minimum pulse height of 1 800 V.